Tops Tech MODULE 3 Assessment

CCNA - NETWORK TROUBLESHOOTING

THEME: SMALL STARTUP OFFICE NETWORK

SECTION A: Scenario-Based Concept Application

1. **Explain how Deep Packet Inspection (DPI) at the Network and Transport layers of the OSI model allows a startup to isolate specific traffic bottlenecks causing slow internet speeds despite high available bandwidth.**

Answer:

- DPI (Deep Packet Inspection) is a technique that reads the full contents of data packets — not just headers — to understand what type of traffic is flowing through the network.
- At Network Layer (Layer 3): DPI inspects source/destination IP addresses to identify which devices or servers are generating heavy traffic.
- At Transport Layer (Layer 4): DPI checks port numbers and protocols (TCP/UDP) to identify specific applications — e.g., video streaming on port 443, or file downloads on port 80.
- How it helps find bottlenecks: Even if total bandwidth is high, one application (like a backup tool or video call) may be consuming most of it. DPI reveals exactly which IP/port is causing congestion.
- Example: DPI may show that one PC is uploading large files via FTP (port 21), hogging the bandwidth — causing slow internet for everyone else, despite high available bandwidth on paper.

2. **Explain the distinct roles of DHCP and DNS in a scenario where startup devices are successfully assigned IP addresses but remain unable to resolve external web addresses for internet access.**

Answer:

- DHCP (Dynamic Host Configuration Protocol): Automatically assigns IP address, subnet mask, default gateway, and DNS server address to devices. If devices have IP addresses, DHCP is working correctly.
- DNS (Domain Name System): Translates domain names (like www.google.com) into IP addresses. Without DNS, the browser cannot find where to go, even though the device is connected.
- In this scenario: DHCP is fine (IPs assigned). DNS is the problem — either the wrong DNS server IP was given by DHCP, or the DNS server is unreachable/down.
- How to verify: Try pinging a known IP (e.g., ping 8.8.8.8). If it works, internet connectivity is fine, but DNS is broken. If it fails, it is a routing or gateway issue.
- Fix: Set the correct DNS server (e.g., 8.8.8.8 for Google DNS) in the DHCP pool or manually on the device.

3. Explain the necessity of Inter-VLAN Routing and the specific role of static routes when a startup requires communication between two isolated departments on the same physical switch.

Answer:

- VLANs (Virtual LANs) logically separate departments on the same physical switch. Devices in VLAN 10 (HR) cannot talk to devices in VLAN 20 (IT) by default — this is the purpose of VLANs (isolation).
- Inter-VLAN Routing is needed: To allow communication between VLANs, traffic must pass through a Layer 3 device (router or Layer 3 switch). This is called Inter-VLAN Routing.
- Two methods: (1) Router-on-a-Stick — one router with sub-interfaces for each VLAN; (2) Layer 3 Switch with SVI (Switched Virtual Interfaces).

- Role of Static Routes: When there is no dynamic routing protocol, static routes are manually configured to tell the router: 'To reach the VLAN 20 network, send traffic via this interface/gateway.' This provides controlled, predictable routing between departments.
- Example: ip route 192.168.20.0 255.255.255.0 192.168.10.1 — tells the router the path to VLAN 20.

4. **Explain the performance differences between ICMP and HTTP/HTTPS protocols that would cause a website to experience high latency while a ping test returns a fast, successful response.**

<u>Answer:</u>

- ICMP (ping): A very lightweight protocol. It just sends a small echo request and gets an echo reply. No connection setup, no data transfer, no encryption — very fast.
- HTTP/HTTPS: Much more complex. It involves TCP 3-way handshake, DNS resolution, SSL/TLS encryption handshake (for HTTPS), server processing time, and data transfer — all before the page loads.
- Why is ping fast but the website is slow:
- • The network path may be fine (ICMP works), but the web server is slow or overloaded.
- • SSL/TLS handshake adds extra round-trips, increasing latency.
- • Firewalls or proxies may inspect/delay HTTP traffic but let ICMP pass freely.
- • DNS resolution may be slow, adding delay before the HTTP connection even starts.
- Conclusion: Ping only tests basic connectivity. HTTP/HTTPS tests the full application stack — a fast ping does NOT guarantee a fast website.

5. **Explain the conceptual logic of Firewall Access Control Lists (ACLs) and the strategic placement of a firewall to restrict unauthorized traffic between a startup's management and general staff departments.**

Answer:

- ACL (Access Control List): A set of rules that permit or deny traffic based on criteria like source IP, destination IP, protocol, or port number. Rules are checked top-to-bottom; first match wins.
- Conceptual Logic: 'If traffic matches this rule — ALLOW or DENY it. If nothing matches — apply default rule (usually DENY ALL).'
- Example ACL rule: Deny all traffic from 192.168.30.0/24 (Staff VLAN) to 192.168.10.0/24 (Management VLAN).
- Strategic Firewall Placement:
- • Place the firewall between the Management VLAN and the Staff VLAN (inter-VLAN boundary).
- • Apply inbound ACL on the Staff-facing interface to block unauthorized access to the Management subnet.
- • Allow only necessary traffic (e.g., allow DNS and internet, but deny direct access to management servers).
- Best Practice: Apply ACL as close to the source (Staff side) as possible to block bad traffic early and reduce load on the network.

SECTION B: Practical Application

Task 1: Design a VLSM subnet scheme for 3 departments requiring 15 hosts each using 192.168.10.0/24.

Answer:

- For 15 hosts per department: We need at least 15 + 2 = 17 addresses (network + broadcast). Next power of 2 = 32. So use /27 (32 addresses, 30 usable hosts). This satisfies 15 hosts easily.

- Department 1 (e.g., HR):
- Network: 192.168.10.0 /27
- Range: 192.168.10.1 – 192.168.10.30
- Broadcast: 192.168.10.31
- Subnet Mask: 255.255.255.224
- Department 2 (e.g., IT):
- Network: 192.168.10.32 /27
- Range: 192.168.10.33 – 192.168.10.62
- Broadcast: 192.168.10.63
- Subnet Mask: 255.255.255.224
- Department 3 (e.g., Finance):
- Network: 192.168.10.64 /27
- Range: 192.168.10.65 – 192.168.10.94
- Broadcast: 192.168.10.95
- Subnet Mask: 255.255.255.224

Task 2: Configure VLANs 10, 20, and 30 on a Cisco switch and assign physical interfaces to each.

Answer:
Cisco Switch CLI Commands:
Step 1 – Create VLANs:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name IT
Switch(config-vlan)# exit
Switch(config)# vlan 30
```

```
Switch(config-vlan)# name Finance
Switch(config-vlan)# exit
```

Step 2 – Assign interfaces to VLANs:

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
```

Task 3: Set up a DHCP pool on a Router/L3 switch to provide dynamic addressing for the VLAN 10 subnet.

Answer:

Router CLI Commands for DHCP Pool (VLAN 10 = 192.168.10.0/27):

```
Router> enable
Router# configure terminal
Router(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.5
Router(config)# ip dhcp pool VLAN10_POOL
Router(dhcp-config)# network 192.168.10.0 255.255.255.224
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

- excluded-address: Reserves IPs for router, servers, printers (they get static IPs).
- network: Defines the subnet the DHCP pool serves.
- default-router: Gateway IP given to DHCP clients.
- dns-server: DNS server IP given to clients for name resolution.

Task 4: Capture a network trace in Wireshark and identify the IP addresses in the DNS query and HTTP GET headers.

Answer:

- Step 1: Open Wireshark, select the active network interface (e.g., Ethernet or Wi-Fi), and click the blue shark fin to start capture.
- Step 2: Open a browser and visit a website (e.g., http://example.com).
- Step 3: Stop capture after a few seconds.
- Step 4 – Find DNS Query: In the filter bar, type: dns — Look for a packet with 'Standard query A example.com'. In the packet details: Source IP = your PC's IP, Destination IP = DNS server IP (e.g., 8.8.8.8).
- Step 5 – Find HTTP GET: In the filter bar, type: http — Look for a packet labeled 'GET / HTTP/1.1'. In the packet details: Source IP = your PC's IP, Destination IP = web server's IP. The Host field in the HTTP header shows the domain name.
- Success Criteria: DNS query and HTTP GET packets are visible, and IP addresses are identifiable in the packet details pane.

SECTION C: Integrated Case Study

Scenario:

TechLaunch, a startup with 40 employees, is experiencing 'flapping' network connectivity. Employees in the Marketing wing report that their internet drops for 30 seconds every few minutes. The Engineering wing, located on a different VLAN, remains perfectly stable. A ping test from a Marketing PC to the Core Switch shows a consistent 25% packet loss and fluctuating latency.

C-Q1. Based on the symptoms, identify the two most likely OSI layers where this fault is occurring.

Answer:

- Layer 1 – Physical Layer: 25% packet loss and flapping connectivity strongly suggest a physical issue — a faulty cable, loose connector, or damaged network port in the Marketing wing.
- Layer 2 – Data Link Layer: 'Flapping' (connection dropping and reconnecting) is a classic Layer 2 symptom. It could indicate a switching loop, MAC address table instability, or a misconfigured port causing broadcast storms on the Marketing VLAN.
- Key clue: The Engineering VLAN is stable, which means the core network is fine. The problem is isolated to Marketing's Layer 1/Layer 2 segment.

C-Q2. Outline a 4-step 'Bottom-Up' troubleshooting sequence to isolate this specific issue.

Answer:

Bottom-Up = Start from Layer 1 (Physical) and move up:

- Step 1 – Layer 1 (Physical): Check all cables in the Marketing wing. Look for damaged, loose, or incorrectly connected cables. Try replacing the patch cable from the Marketing PC to the switch. Check switch port LEDs — amber = error, green = good.
- Step 2 – Layer 2 (Data Link): Log into the switch and check: show interfaces for errors, show mac address-table for instability, show spanning-tree for loop detection. Look for high collision counts or CRC errors.
- Step 3 – Layer 3 (Network): Ping the default gateway from a Marketing PC. Check if IP addressing is correct. Verify the Marketing VLAN's SVI (gateway IP) on the switch is up/up.

- Step 4 – Layer 4+ (Transport/Application): If Layers 1–3 are fine, test TCP connectivity (telnet or browser). Check for DHCP lease issues or ACLs blocking traffic. Run Wireshark to capture traffic and look for anomalies.

C-Q3. List three CLI commands you would run on the switch to check for cable faults or broadcast storms.

Answer:

- Command 1: show interfaces fa0/X — Shows input/output errors, CRC errors, collisions, and interface status. High error counts = cable fault or duplex mismatch.

```
Switch# show interfaces fa0/1
```

- Command 2: show interfaces fa0/X counters — Shows packet counts, broadcast/multicast counts. High broadcast count = possible broadcast storm on that port.

```
Switch# show interfaces fa0/1 counters
```

- Command 3: show spanning-tree vlan 10 — Shows STP topology. If a port is in the wrong state (Forwarding when it should be blocking), a switching loop may exist.

```
Switch# show spanning-tree vlan 10
```

C-Q4. If the issue is a 'Switching Loop,' explain which protocol should have prevented this and why it might have failed.

Answer:

- Protocol: STP — Spanning Tree Protocol (IEEE 802.1D) or its faster version, RSTP (802.1w).

- What STP does: STP automatically detects loops in the network by electing a Root Bridge and placing redundant ports into a Blocking state so that only one active path exists at a time.
- Why STP might have failed:
- • STP was disabled on the VLAN or the switch port.
- • PortFast was enabled on a trunk port (PortFast should only be on access/end-device ports). This can cause loops when a switch is accidentally connected.
- • A new unmanaged switch was added by a user in the Marketing wing, creating a loop without STP awareness.
- • BPDU Guard was not enabled — so a rogue switch could connect and disrupt STP topology.
- Fix: Enable RSTP on all VLANs, enable BPDU Guard on access ports to shut them down if a switch is plugged in unexpectedly.

C-Q5. Propose a long-term configuration change to ensure an error in one department doesn't impact the rest of the office.

Answer:

- 1. Strict VLAN Segmentation: Keep each department on its own VLAN (Marketing = VLAN 10, Engineering = VLAN 20, etc.). This ensures a broadcast storm or loop in one VLAN does not flood other VLANs.
- 2. Enable BPDU Guard + PortFast on all access ports: Prevents rogue switches or loops from being accidentally introduced. If a switch is plugged into an access port, BPDU Guard shuts it down immediately.

```
Switch(config-if)# spanning-tree portfast
Switch(config-if)# spanning-tree bpduguard enable
```

- 3. Enable Storm Control: Limits the rate of broadcast/multicast/unknown unicast traffic on each port. If a broadcast storm starts, the port is automatically throttled or shut down.

```
Switch(config-if)# storm-control broadcast level 20
```

- 4. Use RSTP (Rapid Spanning Tree): Replace old STP with RSTP for faster convergence (convergence in ~1–2 seconds vs 30–50 seconds with STP), minimising the impact of topology changes.
- 5. Monitor with SNMP/Syslog: Set up centralized logging so any port flap, error, or storm triggers an alert to the network admin before it becomes a major outage.